

NEW ESTIMATORS FOR PARETO PARAMETERS BASED ON FREQUENCY MOMENTS

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Summary: A new estimator of the shape parameter of a Pareto distribution, based on the frequency moment estimation principle propounded in Manas (1997) is suggested. The exact distribution of the estimator is established for all sample sizes and the mean and variance are deduced in terms of the generalised Zeta function. Asymptotic expansions of moments of the estimator are obtained, from which an estimator with smaller bias is deduced, and a theoretical verification of the full efficiency of the estimator is derived.

1. Introduction

Manas (1997) has outlined a frequency moment approach to parameter estimation based on some earlier work of Sichel (1947, 1949, 1982). This procedure suggests using the sample geometric mean (G) to estimate scale in a Pareto distribution of the first kind (i.e. the Pareto (I) distribution as defined in Arnold (1983, p.44)). In order to estimate both parameters, an additional statistic is sought which imparts information about location. Equations for the expectations of the two statistics are then solved simultaneously to derive

the estimators. It is obviously advisable to choose an additional statistic with relatively low variance and covariance. Quandt (1966) uses the arithmetic mean to estimate location. In this paper, the minimum order statistic of a sample ($X_{(1)}$) is employed and yields more efficient estimators because its variance and its covariance with G are both of a lower order of magnitude than those of $\bar{X}$.

A Pareto population of the first kind has density function

$$f(x; k, a) = \frac{ak^a}{x^{a+1}}, \quad (x \geq k > 0, a > 0) \quad (1.1)$$

where a is known as the *shape* parameter and k the *threshold* parameter. Quandt (1966) considers a method of estimation of a and k which consists of simultaneously solving equations for the expectations of the sample mean and the minimum order statistic. This gives the estimators

$$\hat{a} = \frac{n\bar{X} - X_{(1)}}{n\{\bar{X} - X_{(1)}\}} \quad (1.2)$$

and

$$\hat{k} = \left(1 - \frac{1}{\hat{a}n}\right) X_{(1)}, \quad (1.3)$$

where n denotes the size of a random sample, $X_1, \dots, X_n$ from (1.1). Quandt shows that these estimators are consistent. In addition, computer simulations reveal that the estimator is biased, and for large n is less efficient by a factor of at least 2 than its maximum likelihood counterpart, $\hat{a}_{ML} = n / \sum_i \log(X_i / \hat{k}_{ML})$, where $\hat{k}_{ML} = X_{(1)}$, (Arnold, 1983, p. 194).

In the present work, it is shown that by replacing the arithmetic by the geometric mean in the above procedure, the estimates obtained are fully efficient and far more amenable to mathematical analysis. Section 2 deals with

the derivation of the estimator and its probability distribution. In Section 3, the moments of the estimator are examined. The exact mean and variance are obtained in terms of the generalised Zeta function and evaluated for various values of a and n . In Section 4, asymptotic expansions of the first four moments of the estimator are given. Based on one of the expansions, an improved estimator is suggested, which is more efficient and less biased. The final section contains a discussion of the results.

2. The estimator of a and its density function

From Manas (1997), an estimating equation derived from frequency moments for estimating k and a is given by

$$G = \hat{k} \left(1 - \frac{1}{\hat{a}n} \right)^{-n}, \quad (2.1)$$

where G denotes the sample geometric mean (i.e. $G = (\prod_i X_i)^{1/n}$). In order to estimate both parameters, an additional relationship between sample values and population parameters is required. The chosen sample value here is the minimum order statistic, $X_{(1)}$, which has probability distribution

$$F(x_{(1)}) = 1 - \left(\frac{k}{x_{(1)}} \right)^{an}, \quad (x \geq k > 0, a > 0) \quad (2.2)$$

i.e. $X_{(1)}$ has a Pareto distribution with threshold parameter k and shape parameter an .

Since $X_{(1)}$ has mean $kan/(an - 1)$, moment estimators of the parameters based on $X_{(1)}$ must satisfy

$$X_{(1)} = \hat{k} \left(1 - \frac{1}{\hat{a}n} \right)^{-1}. \quad (2.3)$$

From (2.1) and (2.3) it follows that the estimator of a is calculated from

$$Y = \frac{X_{(1)}}{G} = \left(1 - \frac{1}{\hat{a}n}\right)^{n-1}, \quad (2.4)$$

or equivalently

$$\hat{a} = \frac{1}{n \{1 - Y^{1/(n-1)}\}}, \quad (2.5)$$

where Y is the ratio of the minimum order statistic to the geometric mean of the sample. Note that if $\hat{a}$ has been evaluated, either (2.1) or (2.3) may be used to estimate k .

We derive the density of $\hat{a} = Z$ by establishing the joint density of $(G, X_{(1)})$, and hence Y . To obtain the joint distribution, we consider the conditional distribution of G given $X_{(1)}$. The (unconditional) density of G can easily be obtained by noting that for each $i = 1, \dots, n$, $\log(X_i/k)$ has an exponential distribution with mean $1/a$, so that $\frac{1}{n} \sum_{i=1}^n \log X_i = \log G$ has a (shifted) gamma distribution with parameters an , a and location (shift) parameter $\log k$. Hence the density of G is found to be given by

$$f(g) = \frac{(an)^n}{k\Gamma(n)} \left(\frac{k}{g}\right)^{an+1} \left\{\log\left(\frac{g}{k}\right)\right\}^{n-1} \quad (g \geq k), \quad (2.6)$$

from which it may be deduced that the t -th moment of G is $E(G^t) = k^t(1 - t/\{an\})^{-n}$, indicating that the frequency-moment estimator, (2.1) is also, in addition, the first moment estimator based on G . (The t -moment can be more easily evaluated using $E(G^t) = \{E(X^{t/n})\}^n$.) In addition, it follows from (2.6) that the density function of the product of the sample observations, $W = G^n$, is given by

$$f(w) = \frac{(ak^a)^n}{\Gamma(n)} w^{-a-1} \left\{\log\left(\frac{w}{k^n}\right)\right\}^{n-1} \quad (w \geq k^n). \quad (2.7)$$

Now, to obtain the conditional distribution of G given $X_{(1)}$, we suppose, without loss of generality, that $X_{(1)} = X_1$. Clearly,

$$\begin{aligned} F(g|x_1) &= P\{G \leq g | X_{(1)} = x_1\} \\ &= P\{\Pi_{i=2}^n X_i \leq g^n/x_1 | X_i \geq x_1, i = 2, \dots, n\}, \end{aligned} \quad (2.8)$$

where the conditional distribution of each X_i ($i = 2, \dots, n$), given that it exceeds x_1 , is Pareto with threshold parameter x_1 and shape parameter a . Thus, by (2.8), the required conditional distribution may be obtained by considering the product of $n-1$ independent Pareto variables, each with threshold parameter x_1 and shape parameter a . In particular,

$$F(g|x_1) = \int_{x_1^{n-1}}^{g^n/x_1} f(w')dw', \quad (2.9)$$

where $f(w')$ is found by replacing n by $n-1$ and k by x_1 in (2.7). Finding the derivative of (2.9) with respect to g gives

$$f(g|x_{(1)}) = \frac{(an)^{n-1}}{\Gamma(n-1)} x_{(1)}^{an} g^{-an-1} \left\{ \log \left(\frac{g}{x_{(1)}} \right) \right\}^{n-2} \quad (g \geq x_{(1)}). \quad (2.10)$$

The joint density of the 2-dimensional random vector $(G, X_{(1)})$ is thus found from $f(g, x_{(1)}) = f(g|x_{(1)})f(x_{(1)})$, where $f(x_{(1)}) = F'(x_{(1)})$ with F given in (2.2), to be given by

$$f(g, x_{(1)}) = \frac{(ank^a)^n}{\Gamma(n-1)} x_{(1)}^{-1} g^{-an-1} \left\{ \log \left(\frac{g}{x_{(1)}} \right) \right\}^{n-2} \quad (g \geq x_{(1)} \geq k). \quad (2.11)$$

From (2.11) the expectation of the product, $GX_{(1)}$, is given by

$$E(GX_{(1)}) = \frac{(an)^n k^2}{(an-1)^{n-1}(an-2)}, \quad (2.12)$$

provided that $a \geq 2/n$. The covariance of G and $X_{(1)}$ is thus

$$\begin{aligned} \text{Cov}(G, X_{(1)}) &= \left(1 - \frac{1}{an}\right)^{-n} \frac{k^2}{(an-2)(an-1)} \\ &= \left(\frac{k}{a}\right)^2 \frac{e^{1/a}}{n^2} + O(n^{-3}), \end{aligned} \quad (2.13)$$

which is an order of magnitude lower than $\text{Cov}(G, \bar{X}) = \left(\frac{k}{a-1}\right)^2 e^{1/a}/n + O(n^{-2})$.

From (2.11), the density function of Y is easily found to be

$$f(y) = \frac{(an)^{n-1}}{\Gamma(n-1)} y^{an-1} \{-\log y\}^{n-2} \quad (0 \leq y \leq 1), \quad (2.14)$$

using a bivariate transformation with $y = x_{(1)}/g$ and $u = x_{(1)}$, say, from which it follows that Y and $X_{(1)}$ are independent. From (2.14) it follows that $-\log Y$ has a gamma distribution with parameters $n-1$ and an . The maximum likelihood estimator of a is thus distributed as the reciprocal of a gamma variate. From (2.14), the density function of $\hat{a}$ is found from the univariate transform, $z = [n \{1 - y^{1/(n-1)}\}]^{-1}$, to be given by

$$f(z) = \frac{\{an(n-1)\}^{n-1}}{n\Gamma(n-1)} \frac{1}{z^2} \left(1 - \frac{1}{nz}\right)^{an(n-1)-1} \left\{-\log \left(1 - \frac{1}{nz}\right)\right\}^{n-2}, \quad (2.15)$$

over the range $1/n \leq z < \infty$. Figures 1-3 illustrate the density, (2.15), for $a = 1.5; 2.5; 10$ and $n = 4; 10; 20$.

3. Moments of the estimator

The substitution $t = -\log(1 - 1/\{nz\})$ into $\int_{1/n}^{\infty} z^r f(z) dz$, with $f(z)$ as in (2.15), gives the integral representation of the r -th moment about the origin of

$Z = \hat{a}$:

$$E(Z^r) = \frac{\{an(n-1)\}^{n-1}}{n^r \Gamma(n-1)} \int_0^\infty \frac{t^{n-2} e^{-an(n-1)t}}{(1-e^{-t})^r} dt. \quad (3.1)$$

Expanding $(1-e^{-t})^{-r}$ in powers of e^{-t} and integrating the series term by term yields the equivalent series expression

$$E(Z^r) = \frac{1}{n^r} \sum_{m=0}^{\infty} \binom{r+m-1}{r-1} / g(m), \quad (3.2)$$

where $g(m) = [1 + m/\{an(n-1)\}]^{n-1}$.

Now, since

$$\frac{\binom{r+m-1}{r-1}}{g(m)} \sim \frac{\{an(n-1)\}^{n-1}}{\Gamma(r)m^{n-r}} \quad \text{as } m \rightarrow \infty, \quad (3.3)$$

it may be deduced that the r -th moment of $\hat{a}$ exists, provided that $n \geq r+2$. Thus it has finite mean if $n \geq 3$ and finite variance if $n \geq 4$. This can also be obtained from the characteristic function of Z ,

$$\phi(s) = e^{is/n} \left\{ 1 - \frac{is}{n} \sum_{m=1}^{\infty} \frac{1}{mg(m)} L'_m(-is/n) \right\}, \quad (3.4)$$

where $L_m(x)$ is the Laguerre polynomial defined by $(1-t) \sum_{m=0}^{\infty} L_m(x)t^m = e^{-xt/(1-t)}$. This approach, however, does not appear to give any better results than the direct calculation of moments about 0.

Now, the generalised Zeta function (also known as the Hurwitz function) of order ν and parameter θ is defined by the integral transform

$$\zeta(\nu, \theta) = \frac{1}{\Gamma(\nu)} \int_0^\infty \frac{t^{\nu-1} e^{-\theta t}}{1-e^{-t}} dt \quad (\nu > 1, \theta > 0) \quad (3.5)$$

or equivalently, the series

$$\zeta(\nu, \theta) = \sum_{m=0}^{\infty} (\theta + m)^{-\nu} \quad (3.6)$$

(Spanier and Oldham, 1987, p.655).

The expectation of $\hat{a}$ is thus given by either (3.1) or (3.2) with $r = 1$ and is thus

$$E(Z) = \frac{c_n^{n-1}}{n} \zeta(n-1, c_n) = \frac{1}{n} \sum_{m=0}^{\infty} \left\{ 1 + \frac{m}{c_n} \right\}^{-n+1}, \quad (3.7)$$

where $c_n = an(n-1)$.

The second moment of Z about the origin is given by either (3.1) or (3.2) when $r = 2$ provided that $n \geq 4$. Integrals of the form (3.1) for $r = 2$ are related to the generalised Zeta function by the identity

$$\frac{1}{\Gamma(\nu)} \int_0^{\infty} \frac{t^{\nu-1} e^{-\theta t}}{(1-e^{-t})^2} dt = \zeta(\nu-1, \theta-1) - (\theta-1)\zeta(\nu, \theta-1) \quad (3.8)$$

(Gradshteyn and Ryzhik, 1980; 2, §3.423(4); p.330). The second moment is thus

$$E(Z^2) = \frac{c_n^{n-1}}{n^2} \{ \zeta(n-2, c_n-1) - (c_n-1)\zeta(n-1, c_n-1) \}, \quad (3.9)$$

from which the variance of Z may be deduced using (3.7) and (3.9). Values of $Var(Z)$ are provided in table 1 for various values of a and n . Higher order moments may also be deduced in terms of the generalised Zeta function using repeated integration by parts of (3.1).

4. Asymptotic expansions of moments and an improved estimator

Asymptotic expansions (up to terms of order $O(n^{-3})$) in ascending powers of n^{-1} may be derived for the first four moments of Z using the Euler-Maclaurin summation formula (Milne-Thomson, 1933; 4, §(4); p.189) and are given by

$$E(Z) = a + \frac{2a+1}{2n} + \frac{24a+a^{-1}}{12n^2} + O(n^{-3}) \quad (4.1)$$

$$E(Z^2) = a^2 + \frac{a(3a+1)}{n} + \frac{120a^2 + 12a + 5}{12n^2} + O(n^{-3}) \quad (4.2)$$

$$E(Z^3) = a^3 + \frac{3a^2(4a+1)}{2n} + \frac{a(62a^2 + 9a + 2)}{2n^2} + O(n^{-3}) \quad (4.3)$$

$$E(Z^4) = a^4 + \frac{2a^3(5a+1)}{n} + \frac{a^2(450a^2 + 72a + 11)}{6n^2} + O(n^{-3}). \quad (4.4)$$

Details of the calculations are available from the authors on demand. From the above, asymptotic expansions of the corresponding central moments are found to be

$$Var(Z) = E\{Z - E(Z)\}^2 = \frac{a^2}{n} \left(1 + \frac{5}{n}\right) + O(n^{-3}) \quad (4.5)$$

$$E\{Z - E(Z)\}^3 = O(n^{-2}) \quad (4.6)$$

$$E\{Z - E(Z)\}^4 = \frac{3a^4}{n^2} + O(n^{-3}). \quad (4.7)$$

From (4.1) it is obvious that Z is asymptotically unbiased for a . Also, from (4.5), it is clear that Z is fully efficient for sufficiently large n . The Cramér-Rao lower bound for the variance of estimators of a is a^2/n and the variance of the maximum likelihood estimator is quoted for comparison:

$$Var(\hat{a}_{ML}) = \frac{a^2 n^2}{(n-2)^2(n-3)} = \frac{a^2}{n} \left(1 + \frac{7}{n}\right) + O(n^{-3}) \quad (4.8)$$

(Arnold, 1983, p.195). The estimator, Z , thus has lower variance (up to terms of $O(n^{-3})$) than $\hat{a}_{ML}$ (see Table 1).

If Z is standardised by putting $W = (Z - a)/(a/\sqrt{n})$, then the asymptotic expansions of the moments of W are given by

$$E(W) = \frac{2a+1}{2a\sqrt{n}} + O(n^{-3/2}) \quad (4.9)$$

$$Var(W) = E\{W - E(W)\}^2 = 1 + \frac{5}{n} + O(n^{-2}) \quad (4.10)$$

$$E\{W - E(W)\}^3 = O(n^{-1/2}) \quad (4.11)$$

$$E\{W - E(W)\}^4 = 3 + O(n^{-1}). \quad (4.12)$$

From (4.9)–(4.12), it is clear that the asymptotic values of the first four moments are identical with those of a standardised normal random variable. It thus seems reasonable to conjecture that the distribution of Z approaches normality as $n \rightarrow \infty$. From (4.1)–(4.4), it would also seem reasonable to conjecture that for any $r = 1, 2, \dots$

$$E(Z^r) = a^r + \frac{ra^{r-1}}{2n} \{1 + (r+1)a\} + O(n^{-2}). \quad (4.13)$$

Now, the expansion given in (4.1) for the mean of $\hat{a}$ reveals that the estimator has bias of order $O(n^{-1})$. In fact, $E(\hat{a}) = a(n+1)/n + (2n)^{-1} + O(n^{-2})$, from which the improved estimator,

$$\hat{a}^* = \frac{n}{n+1} \left(Z - \frac{1}{2n} \right), \quad (4.14)$$

may be deduced, which has bias of order $O(n^{-2})$. It is clear that the transformation of Z in (4.14) also reduces variance, particularly for small sample sizes. In table 1, numerical values are given for the means and variances of $\hat{a}$, $\hat{a}^*$, $\hat{a}_{ML}$ and $\hat{a}_{ML}^*$, where $\hat{a}_{ML}^* = (n-2)\hat{a}_{ML}/n$ denotes the estimator derived by correcting $\hat{a}_{ML}$ for bias (Arnold, 1983, p. 195). The performance of the new estimator (in terms of bias and variance) is compared with that of $\hat{a}_{ML}^*$.

5. Discussion

The values in Table 1 indicate the rate at which $\hat{a}$ becomes fully efficient for increasing n . The column marked % *Rel. Eff.* is the improvement in efficiency obtained by using $\hat{a}_{ML}^*$ instead of $\hat{a}^*$ and is calculated from $\{Var(\hat{a}^*)/Var(\hat{a}_{ML}^*) - 1\} \times 100\%$. (In the calculations, many more decimal

places were used than the five quoted.) For the smallest values of n , $\hat{a}_{ML}^*$ is clearly more efficient, though for $n = 20$, $\hat{a}_{ML}^*$ is only about 1% more efficient. This relative efficiency continues to decrease (with increasing n) as steeply as the Cramér–Rao bound permits, so that for $n \geq 35$ the advantages of $\hat{a}_{ML}^*$ become negligible and for large n , $\hat{a}$ shares all the optimal properties of $\hat{a}_{ML}^*$.

It is noteworthy that the variance of $\hat{a}$ (even before bias correction) is uniformly lower than that of the maximum likelihood estimator, $\hat{a}_{ML}$, as is apparent from (4.5) and (4.8). This rate of decrease remains more or less the same for each chosen value of a . On closer inspection of Table 1, we see that for each n , the relative improvement in efficiency obtained by using $\hat{a}_{ML}^*$ increases with increasing a . In particular, the differences in relative efficiency for each n between the cases $a = 0.5$ and $a = 1.5$ are far greater than those between $a = 2.5$ and $a = 10$, which are very close, indicating the lack of dependency on a in this range. Thus for values of a encountered in practice, conservative estimates for the relative efficiency of $\hat{a}^*$ may be obtained from Table 1 in the case $a = 10$.

On the computational side, numerical values of the generalised Zeta function were computed using an algorithm based on Padé approximants as given in Spanier and Oldham (1987, p.658). With order, a , and parameter, $an(n-1)$, this function becomes increasingly difficult to evaluate as n becomes larger and assumes values so minutely small that $n > 30$ would create an underflow on most mainframe computers. From (3.7) and (3.9), we see that these small values are typically multiplied by factors which become very large for increasing n . Fortunately, however, it is not necessary to compute these

for larger n as the rapid rate of convergence to full efficiency soon becomes evident.

Finally, it is worth noting that $\hat{a}$ may be derived from an exponential transformation of $\hat{a}_{ML}$, in particular $Y = \exp\{-1/\hat{a}_{ML}\}$. This suggests equating Y with its expectation, which is quickly found from (2.14) to be $(1 + 1/\{an\})^{-n+1}$. The estimator obtained in this way is

$$\hat{a}^{**} = \frac{1}{n\{Y^{1/(-n+1)} - 1\}} \quad (5.1)$$

The similarity between this and (2.5) is striking. In a subsequent paper, refined estimators for both parameters will be examined based on frequency moments of more general order, of which (2.5) and (5.1) are special cases.

6. References

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Table 1. Means and Variances of $\hat{a}$ and $\hat{a}^*$ and % relative efficiency of $\hat{a}^*$.

$$\alpha = 0.5$$

n	E($\hat{a}$)	E($\hat{a}^*$)	Var($\hat{a}$)	Var($\hat{a}^*$)	Var($\hat{a}_{ML}$)	Var($\hat{a}_{ML}^*$)	Rel. Eff.
4	0.88532	0.60286	0.55741	0.35674	1.00000	0.25000	42.70%
5	0.77330	0.56108	0.22003	0.15280	0.34722	0.12500	22.24%
6	0.71295	0.53967	0.12906	0.09482	0.18750	0.08333	13.79%
7	0.67482	0.52797	0.08933	0.06839	0.12250	0.06250	9.42%
8	0.64843	0.52083	0.06762	0.05343	0.08889	0.05000	6.86%
9	0.62903	0.51614	0.05413	0.04385	0.06888	0.04167	5.23%
10	0.61416	0.51288	0.04499	0.03719	0.05580	0.03571	4.12%
15	0.57253	0.50550	0.02411	0.02119	0.02774	0.02083	1.69%
20	0.55319	0.50304	0.01636	0.01484	0.01816	0.01471	0.92%
25	0.54201	0.50193	0.01236	0.01143	0.01343	0.011364	0.57%
30	0.53471	0.50133	0.00993	0.00930	0.01063	0.00926	0.40%
35	0.52957	0.50097	0.00829	0.00784	0.00879	0.00781	0.29%

$$\alpha = 1.5$$

n	E($\hat{a}$)	E($\hat{a}^*$)	Var($\hat{a}$)	Var($\hat{a}^*$)	Var($\hat{a}_{ML}$)	Var($\hat{a}_{ML}^*$)	Rel. Eff.
4	2.37847	1.80277	5.05731	3.23668	9.00000	2.25000	43.85%
5	2.10222	1.66852	1.99778	1.38735	3.12500	1.12500	23.32%
6	1.95988	1.60847	1.17072	0.86012	1.68750	0.75000	14.68%
7	1.87256	1.57599	0.80932	0.61964	1.10250	0.56250	10.15%
8	1.81337	1.55633	0.61207	0.48361	0.80000	0.45000	7.46%
9	1.77053	1.54347	0.48950	0.39650	0.61990	0.37500	5.73%
10	1.73806	1.53460	0.40660	0.33603	0.50223	0.32143	4.54%
15	1.64897	1.51465	0.21740	0.19107	0.24963	0.1875	1.90%
20	1.60847	1.50807	0.14744	0.13373	0.16340	0.13235	1.04%
25	1.58531	1.50510	0.11135	0.10295	0.12083	0.10227	0.65%
30	1.57030	1.50352	0.08938	0.08371	0.09566	0.08333	0.45%
35	1.55979	1.50257	0.07464	0.07055	0.07909	0.07031	0.33%

$\alpha = 2.5$

n	$E(\hat{\alpha})$	$E(\hat{\alpha}^*)$	$\text{Var}(\hat{\alpha})$	$\text{Var}(\hat{\alpha}^*)$	$\text{Var}(\hat{\alpha}_{ML})$	$\text{Var}(\hat{\alpha}_{ML}^*)$	Rel. Eff.
4	3.87708	3.00167	14.05729	8.99667	25.00000	0.25000	43.95%
5	3.43467	2.77889	5.55334	3.85648	8.68056	3.12500	23.41%
6	3.20926	2.67937	3.25404	2.39073	4.68750	2.08333	14.76%
7	3.07211	2.62559	2.24932	1.72214	3.06250	1.56250	10.22%
8	2.97969	2.59306	1.70095	1.34396	2.22222	1.25000	7.52%
9	2.91311	2.57180	1.36025	1.10180	1.72194	1.04167	5.77%
10	2.86283	2.55213	1.12982	0.93373	1.39509	0.89286	4.58%
15	2.72579	2.52418	0.60399	0.53085	0.69342	0.52083	1.92%
20	2.66397	2.51331	0.40960	0.37152	0.45389	0.36768	1.05%
25	2.62857	2.50841	0.30932	0.28598	0.33565	0.28409	0.67%
30	2.60599	2.50580	0.24830	0.23254	0.26573	0.23148	0.46%
35	2.59007	2.50424	0.20733	0.19597	0.21974	0.19531	0.34%

$\alpha = 10.0$

n	$E(\hat{\alpha})$	$E(\hat{\alpha}^*)$	$\text{Var}(\hat{\alpha})$	$\text{Var}(\hat{\alpha}^*)$	$\text{Var}(\hat{\alpha}_{ML})$	$\text{Var}(\hat{\alpha}_{ML}^*)$	Rel. Eff.
4	15.12552	12.00042	224.99476	143.99665	400.00000	100.00000	44.00%
5	13.43367	11.11139	88.88659	61.72680	138.88889	50.00000	23.45%
6	12.58357	10.71448	52.08210	38.26440	75.00000	33.33333	14.79%
7	12.07160	10.50015	35.99938	27.56203	49.00000	25.00000	10.25%
8	11.72930	10.37049	27.22173	21.50853	35.55556	20.00000	7.54%
9	11.48423	10.28581	21.76788	17.63199	27.55102	16.66666	5.79%
10	11.30008	10.22735	18.08043	14.94251	22.32143	14.28571	4.60%
15	10.80260	10.09619	9.66512	8.49473	11.09467	8.33333	1.94%
20	10.58058	10.05293	6.55391	5.94459	7.26216	5.88235	1.06%
25	10.45480	10.03346	4.94962	4.57620	5.37034	4.54545	0.68%
30	10.37382	10.02305	3.97300	3.72081	4.25170	3.70370	0.46%
35	10.31732	10.01684	3.31724	3.13551	3.51527	3.12500	0.34%

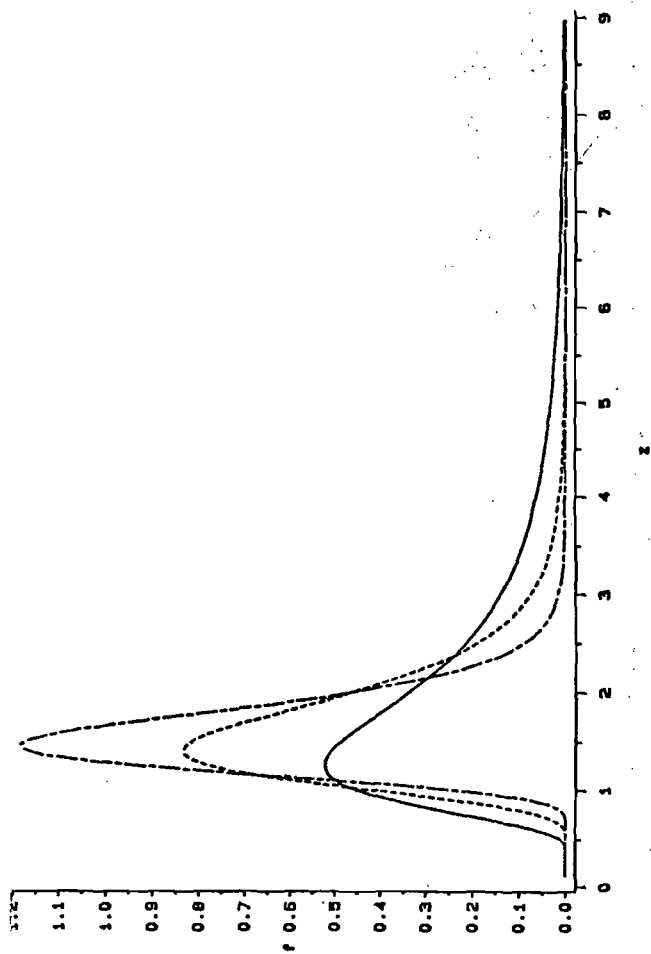


Figure 1.

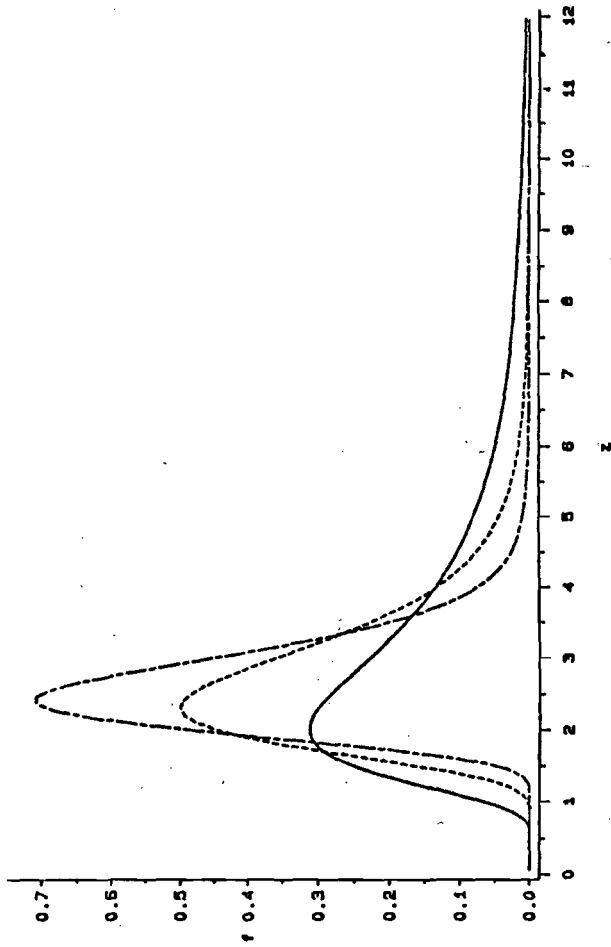


Figure 2.

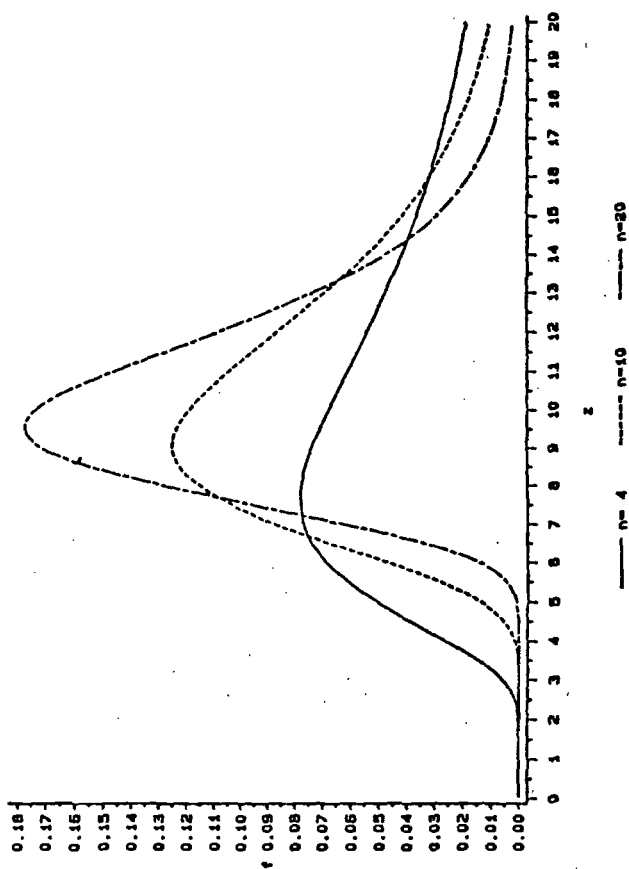


Figure 3.